Unified Financial Analysis

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The Missing Links of Finance

W. Brammertz, I. Akkizidis, W. Breymann,
R. Entin and M. Rüstmann

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Preface

Make everything as simple as possible but not simpler than that.
Albert Einstein

Financial analysis means different things to practitioners across a wide range of industries, disciplines, regulatory authorities and standard setting bodies. In daily practice, an accountant is likely to understand this term as bookkeeping and compliance with accounting standards. To a trader or a quantitative analyst, this term conjures up option pricing, whereas to a treasurer in a bank it can stand for liquidity gap analysis, measurement of market risks and stress testing scenarios. It could mean cost accounting and profitability analysis to a controller or a financial analyst valuing a company. A regulator or a compliance officer using this term has in mind primarily the calculation of regulatory capital charges. On the other hand, a risk officer in an insurance company is likely to think of simulating the one-year distribution of the net equity value using Monte Carlo methods.

The examples mentioned above, and there are many others, make up a vast body of complex and specialized knowledge. It is only natural that practitioners tend to concentrate on a specific topic or subject matter, driven by the necessity of generating a report, an analysis or just a set of figures. The focus on the output drives the tools, systems, methodologies and, above all, the thinking of practitioners. Hedging the gamma risk of an option portfolio, for example, is after all quite different from managing short-term liquidity risk or backtesting a VaR model for regulatory purposes.

This, however, is only superficially true. The mechanisms underlying these disparate analysis types are few, more common, and less complex than one would expect provided that everything is made as simple as possible but not simpler than that. The aim of this book is to expose these shared elements and show how they can be combined to produce the results needed for any type of financial analysis. Our focus lies on the analysis of financial analysis and not a specific analytical need. This also determines what this book is *not* about. While we discuss how market risk factors should be modeled, we cannot hope to scratch the surface of interest rate modeling. To use another example, the intricacies of IFRS accounting as such are not directly relevant to us, but the underlying mechanisms for calculating book values using these rules are.

Why should this be relevant? Given the breadth of specific knowledge required to master these topics, what is the use of analyzing analysis instead of just doing it? The main argument can be seen by looking at the expensively produced mess that characterizes financial analysis in most banks and insurances. Incompatible systems and analysis-centered thinking combine to create cost and quality problems which become especially apparent at the top management level. Capital allocation, to use one example, relies on a combination of risk, income and value figures which are often drawn from different sources and calculated using different assumptions.

One must either invest considerable resources in the reconciliation of these figures or accept analytical results of a lower quality. Both options are becoming less tenable. Reconciling the output of many analytical systems is difficult, error prone and will become more so as analysis grows in sophistication. Regulation, which for better or worse is one of the main drivers of financial analysis, increasingly demands better quality analytical results. Due to the high costs of compliance, regulation is already perceived as a risk source in its own right.[1] In the aftermath of the subprime crises the regulatory burden and the associated compliance costs are bound to increase.

There is a better way to address these pressing issues by approaching financial analysis from a unified perspective. In this book we shall develop an analytical methodology based on well-defined inputs and delivering a small number of analytical outputs that can be used as building blocks for any type of known financial analysis. Our approach is conceptual and we discuss an abstract system or methodology for building up financial analysis. Our thinking about these issues is informed by the practical experience some of us have with implementing such a system in reality. Over a period of 20 years, there has been a reciprocal and symbiotic relationship between the conceptual thinking and its concrete implementation. It started with a doctoral thesis[2] which was soon implemented in banks and a few insurances corroborating the initial ideas. This is why we do not try to build this methodology consistently from first principles, a goal which is anyway not always achievable. Where a choice cannot be traced back to a well-reasoned axiom, we appeal implicitly to our experience with a concrete system implementing this methodology.

This book reflects this background. When using the terms “system” or “methodology” we primarily have in mind a conceptual view of such a system. At the same time, there is also a practical bent to our discussion since there is no value to a conceptual framework if it cannot be implemented in reality. This, however, should not be understood to mean that the target audience is made of system builders or software engineers.

Our target audience are consumers of financial analysis output – the figures and numbers – from a wide range of professional skills and hierarchical levels. The book deals with topics as disparate as IFRS accounting standards, arbitrage-free yield curve modeling using Monte Carlo techniques, option valuation, Basel II and Solvency II regulations, profitability analysis and activity-based costing to name but a few. It is, however, not a book for someone seeking specialized knowledge in any of these fields. Beyond a short introduction to these topics, we assume that the reader has the relevant knowledge or can find it elsewhere. Some help is found at the end of some chapters where a further reading section provides references to relevant literature. Not being a scholarly work, however, we abstain from providing a comprehensive bibliography; in fact we produce none. Rather, what this book attempts to do is to put different analytical concepts in their proper context and show their common underlying principles. Although we hope to satisfy the intellectual curiosity of the reader for topics outside his or

[1] Regulatory risk was top ranked in CSFI surveys of bank risks from 2005 through 2007.

[2] W. Brammertz, *Datengrundlage und Analyseinstrumente für das Risikomanagement eines Finanzinstitutes*, Thesis, University of Zurich, 1991.

her specialized knowledge domain, we believe that the insights gained by a unified approach to financial analysis will also contribute to the understanding of one's specific field.

Part I starts with a short and eclectic history of financial analysis leading to the chaos that characterizes the state of financial analysis today. It is followed by a condensed summary of the main themes of this book. We introduce the principles of unified analysis, and important concepts emerge, such as the natural and investment time horizons, input and analysis elements. We also draw an important distinction between static and dynamic types of analysis. Static analysis or, more accurately, *liquidation view* analysis, is based on the assumption that all current assets and liabilities can be sold at current market conditions without taking the evolution of future business into account. This restriction is removed in the dynamic, or *going-concern*, analysis where new business is also considered.

The concepts introduced in the second chapter are further developed in Part II where input elements are discussed, Part III which deals with analytical elements from a liquidation perspective, and Part IV which addresses them from a going-concern perspective. Finally, Part V demonstrates the completeness of the system, showing that all known types of financial analyses are covered and offering a solution for the current information chaos not only for individual banks and insurances but also from a global – and especially regulatory – perspective.

Having such a wide range of audience and topics, this book will be read differently by different readers:

Students of finance It is assumed that all or most topics covered, such as bookkeeping or financial engineering have already been studied in specialized classes. This book brings together for the reader all the loose threads left by those different classes. It is therefore intended for more advanced students or those with practical experience. Although the more mathematical sections can be safely skipped, the rest of the material is equally relevant.

Senior management Part I, where the main concepts are introduced, and Part V where the main conclusions are drawn, should be of interest. If it is desired to go deeper into details, parts of Chapter 3 and Chapter 4 should be read, choosing the appropriate level of depth.

Practitioners Specialists at different levels in treasury, asset and liability management, risk controlling, regulatory reporting, budgeting and planning and so on should find most parts of this book of interest. Depending on the level within the organization the relevant material may be closer to that of the student or the senior manager. According to one's background and professional focus, some chapters can be read only briefly or skipped altogether. Nevertheless, we believe that at least Parts I and V should be read carefully, as well as Part III which discusses the main building blocks of static financial analysis. In particular, Chapter 3, which discusses financial contracts in detail, should be given due attention because of the centrality of this concept within the analytical methodology. Readers dealing with liquidity management, asset and liability management, planning and budgeting in banks and insurances should find Part IV of special interest. This part could also be of general interest to other readers since dynamic analysis, to our knowledge, is not well covered in the literature.

IT professionals Although this book is not aimed at IT professionals as such, it could nevertheless be interesting to analysts and engineers who are building financial analysis software. The book reveals the underlying logical structure of a financial analysis system and provides a high level blueprint of how such a system should be built.

Financial analysts in nonfinancial industry The book addresses primarily readers with a background in the financial industry. Chapter 17, however, reaches beyond the financial into the nonfinancial sector. Admittedly the path is long and requires covering a lot of ground before getting to the nonfinancial part, but the persevering readers will be able to appreciate how similar the two sectors are from an analytical perspective. In addition to the first part of the book, Part IV should be read carefully in its entirety and in particular Chapter 17 which deals with nonfinancial entities.

Finally, we hope that this book would be of value to any reader with a general interest in finance. Finding the missing links between many subjects which are typically treated in isolation and discovering the common underlying rules of the bewildering phenomena of finance should be worthwhile and enjoyable in its own right.